

E.4 Additional Controls

This section evaluates the robustness of the relationships between employer encouragement of AI chatbots and workers' adoption and reported benefits. We leverage the richness of our data to examine potential confounders on both the firm and worker sides.

On the firm side, we show that all results remain robust when controlling for firm characteristics (Table D.1), ensuring that observed differences across workplaces are not driven by variation in firm age, size, or productivity. On the worker side, we show that the results are similarly robust to controlling for workers' detailed task mixes within occupations, ensuring that the effects of employer encouragement are not merely driven by differences in task types more amenable to AI chatbot use.⁴⁶

Table E.12 and Figure E.20 summarize these robustness checks, showing that our estimates of the impact of employer encouragement on chatbot adoption and perceived benefits remain virtually unchanged after accounting for firm characteristics and worker task mixes.

The robustness of our estimates to the inclusion of this rich set of controls provides reassurance that observable confounders are not driving the results. Moreover, Oster (2019) and Altonji, Elder and Taber (2005) provide conditions under which the stability of coefficients to the inclusion of observable controls can also help rule out selection on *unobservables*.

⁴⁶Task importances are derived from our survey, which asked workers to rate the importance of six representative O*NET job tasks in their occupations; see Appendix J and Humlum and Vestergaard (2025, SI6) for details.

Table E.12: Encouragement, Adoption, and Work (Additional Controls)

(a) Adoption (Figure D.3)								
	Ever Used		Monthly or More		Weekly or More		Daily	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Encouraged	0.365 (0.006)	0.347 (0.006)	0.337 (0.007)	0.321 (0.007)	0.283 (0.006)	0.268 (0.006)	0.136 (0.005)	0.129 (0.005)
Worker characteristics controls	✓	✓	✓	✓	✓	✓	✓	✓
Firm characteristics controls		✓		✓		✓		✓
Worker task mix controls		✓		✓		✓		✓
(b) Reported Benefits (Table D.2)								
	Time Savings		Quality		Creativity		Job Satisfaction	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Encouraged	0.098 (0.008)	0.099 (0.008)	0.115 (0.009)	0.112 (0.009)	0.094 (0.009)	0.094 (0.009)	0.071 (0.007)	0.069 (0.007)
Worker characteristics controls	✓	✓	✓	✓	✓	✓	✓	✓
Firm characteristics controls		✓		✓		✓		✓
Worker task mix controls		✓		✓		✓		✓
(c) New Workloads from AI Chatbots for Adopters (Figure D.5.(a))								
	Same Tasks		New Tasks		Same and New Tasks			
	(1)	(2)	(3)	(4)	(5)	(6)		
Encouraged	0.016 (0.004)	0.016 (0.004)	0.046 (0.005)	0.040 (0.005)	0.008 (0.002)	0.008 (0.002)		
Worker characteristics controls	✓	✓	✓	✓	✓	✓		
Firm characteristics controls		✓		✓		✓		
Worker task mix controls		✓		✓		✓		

Notes: This table presents estimates of the impact of employer encouragement on AI chatbot adoption (Panel a), the reported benefits of adoption (Panel b), the allocation of time savings (Panel c), and new workloads resulting from chatbot use among adopters (Panel d). Odd-numbered columns report our main estimates from Equation (69). Even-numbered columns report specifications where we augment controls for firms' characteristics and workers' detailed task mixes. Robust standard errors are shown in parentheses. *Sample:* All complete responses from the 2024 survey that can be linked to the registry data.

Figure E.20: Employer Encouragement and Worker Gaps in Adoption (Figure E.2)



Notes: This figure illustrates the impact of employer usage policies on worker disparities in AI chatbot adoption. The estimates are obtained from a multivariate regression of AI chatbot adoption on worker characteristics X , controlling for occupation fixed effects, and are estimated separately based on employers' AI chatbot initiatives (Encouraged = 1 vs. Encouraged = 0). Panel (a) presents our main specification from Figure E.2, while Panel (b) adds controls for firm characteristics (Table D.1) and workers' detailed task mixes. Whiskers represent 95% confidence intervals. The reported p-values test whether the coefficients differ between the two groups. *Sample:* All completed responses from the 2024 survey linked to registry data.

E.5 Coworker Encouragement

A key focus of our analysis is the link between employer chatbot initiatives and labor market outcomes. Section 1.4 shows how workers’ adoption and reported work effects vary with these initiatives, and Section 3 uses employer-encouraged use as the main workplace treatment.

While our main analysis relies on self-reported employer encouragement, this appendix demonstrates robustness to using coworker reports. At the individual level, this helps assess whether effects reflect workplace-wide initiatives or individual interventions; at the workplace level, it addresses concerns about attenuation bias from survey measurement error.

We focus the robustness analysis on encouraged use, our primary workplace treatment. Section E.5.1 describes how we construct coworker measures, and Section E.5.2 shows that our findings hold when using coworker encouragement, replicating first the adoption and self-reported work effects from Section 1.4, then the labor market results from Section 3.

E.5.1 Empirical Strategy

To operationalize these ideas, we construct leave-one-out averages of coworkers’ reported employer encouragement for each worker i :

$$\text{Encouraged}_{j,-i} = \frac{1}{N_{j(i)} - 1} \sum_{k \in j(i) \setminus i} \text{Encouraged}_k, \quad (70)$$

where $N_{j(i)}$ denotes the number of respondents in the same workplace-occupation cell $j(i)$ as worker i .

The coworker measurements are made possible by our workplace-based sampling design (described in Section 1.3), which ensures that most respondents have coworkers who also participated in the survey. To mitigate measurement error due to incomplete sample coverage, we apply an empirical shrinkage procedure to the leave-out encouragement rates, $\text{Encouraged}_{j,-i}$ (see Appendix H.2 for details on this method). Importantly, our results in Section E.5.2 are robust to using the raw leave-out means instead.